Deepfake Video Detection for Low-Resolution Video KYC in Financial Systems Using rPPG and Hybrid Quantum Machine Learning

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***Abstract*—The rapid advancement of generative artificial intelligence has facilitated the proliferation of hyper-realistic deepfakes, posing a critical threat to digital trust within financial Video Know Your Customer protocols. Financial institutions have witnessed a 2,137 percent surge in deepfake fraud attempts, highlighting an urgent need for robust authentication mechanisms capable of operating in real-world conditions. This review evaluates the current state of deepfake detection with a specific focus on achieving resilience in low-resolution video streams, where traditional forensic cues are frequently obscured by compression noise and bandwidth limitations. By surveying eight pivotal research papers, this work analyzes the methodological transition from classical Convolutional Neural Networks to advanced physiological monitoring via Remote Photoplethysmography and Hybrid Quantum Machine Learning paradigms. While rPPG-based physiological watermarking offers a novel defense by tracking biological rhythms, significant research gaps persist regarding its sensitivity to environmental lighting and motion artifacts in noisy KYC environments. Furthermore, current quantum frameworks often lack rigorous testing on low-resolution financial datasets. Our findings suggest that the integration of quantum-inspired optimization and attention mechanisms provides a promising trajectory for enhancing feature representation in degraded video. Future directions emphasize the development of confidence-aware, hybrid quantum architectures to bridge the gap between academic detectors and the stringent security requirements of operational financial authentication.**

***Index Terms*—Deepfake Detection, Hybrid Models, Quantum Machine Learning, Low-Resolution Video, rPPG, Video KYC**

I. Introduction

A. Background

Deepfakes are AI-generated synthetic media that manipulate facial appearance using deep learning techniques. Their rise has created serious concerns in identity verification systems.

B. Problem Statement

Detecting deepfake videos in low-resolution KYC environments is challenging due to poor quality, compression artifacts, and lack of reliable detection features.

C. Objectives

The objective of this study is to analyze existing techniques, identify their limitations, and propose a robust detection framework.

D. Organization

The paper is organized as follows: Section II presents the literature review methodology, Section III discusses existing techniques, Section IV provides comparative analysis, Section V describes the proposed framework, and Sections VI–VIII cover challenges, future scope, and conclusion.

II. Review Methodology

This review paper presents a comprehensive analysis of recent advancements in deepfake detection techniques, with a focus on hybrid, multimodal, and physiological-signal-based approaches. A total of eight research papers were selected for the study based on their relevance, recency, and contribution to improving detection accuracy.

The papers were collected from reputed academic databases including IEEE Xplore, Springer, Scopus, and Google Scholar. The selection criteria emphasized works that proposed novel architectures such as convolutional neural networks (CNNs), vision transformers (ViTs), quantum-enhanced models, and rPPG-based detection mechanisms.

Inclusion criteria for the survey included:

- Research focused on deepfake detection
- Papers proposing innovative or hybrid methodologies
- Studies with experimental validation

Exclusion criteria included:

- Papers lacking experimental results
- Studies not directly related to deepfake detection
- Redundant or outdated approaches

The selected works were analyzed based on methodology, results, and limitations to identify current trends and research gaps in the domain.

III. Existing Techniques

To better understand the advancements in deepfake detection, the surveyed techniques are categorized into three major groups based on their underlying methodology.

A. Spatial-Temporal Deep Learning Approaches

These approaches primarily rely on convolutional neural networks (CNNs) combined with temporal models such as recurrent neural networks (RNNs) or gated recurrent units (GRUs) to capture both spatial and sequential features in video data.

The work by Talwalkar et al. [4] utilizes a hybrid CNN-GRU architecture, where CNN models such as DenseNet121 and Xception extract spatial features from video frames, while GRU captures temporal dependencies across frames. This approach significantly improves detection accuracy by learning sequential inconsistencies in manipulated videos.

Similarly, traditional CNN-based models such as MobileNet [8] focus on extracting visual artifacts from images but lack temporal awareness, limiting their robustness in detecting advanced deepfakes.

B. Physiological Signal-Based Approaches

These methods exploit biological signals, particularly remote photoplethysmography (rPPG), which captures subtle changes in skin color caused by blood flow.

Kammari et al. [2] demonstrate that deepfake videos often fail to replicate consistent physiological patterns such as heartbeat rhythms. By extracting rPPG signals from facial regions and analyzing them using machine learning techniques, these methods achieve reliable detection performance.

Susi et al. [5] further enhance this approach by synchronizing physiological signals with deep learning models, improving detection accuracy. However, these methods are sensitive to environmental factors such as lighting and motion, which can affect signal quality.

C. Multimodal and Transformer-Based Approaches

Recent advancements focus on combining multiple modalities such as visual, audio, and physiological data to improve detection accuracy.

Xiang Li et al. [3] propose a multimodal framework that integrates audio and visual features using Vision Transformers and temporal attention mechanisms. This approach captures inconsistencies not only in visual content but also in speech patterns, achieving near-perfect accuracy.

Similarly, Rao et al. [7] utilize multimodal Vision Transformers to process spatial, spectral, and physiological features simultaneously, significantly enhancing detection robustness.

Additionally, emerging research such as Dass et al. [1] and Khan et al. [6] explores quantum-enhanced deep learning models, aiming to further improve classification capabilities. However, these methods are still in experimental stages and rely on simulated environments.

IV. Comparative Analysis

The reviewed deepfake detection techniques exhibit significant variations in terms of methodology, performance, and computational complexity. Traditional CNN-based approaches provide a strong baseline but often lack robustness when dealing with temporal inconsistencies and real-world variations.

Hybrid spatial-temporal models, such as CNN-GRU architectures, demonstrate improved performance by capturing both frame-level and sequence-level features. However, their performance gains are not always consistent across different datasets.

Physiological signal-based approaches, particularly those using rPPG, offer a unique advantage by leveraging biological inconsistencies in deepfake videos. These methods show better generalization but are sensitive to environmental conditions such as lighting and motion.

Multimodal and transformer-based approaches achieve the highest accuracy by integrating multiple data sources such as audio and visual features. Despite their superior performance, they suffer from high computational complexity and limited real-time applicability.

V. Proposed Framework

The proposed system is designed as a two-stage deepfake detection pipeline specifically optimized for low-resolution KYC video environments. The framework integrates physiological signal analysis with hybrid quantum machine learning to improve detection robustness.

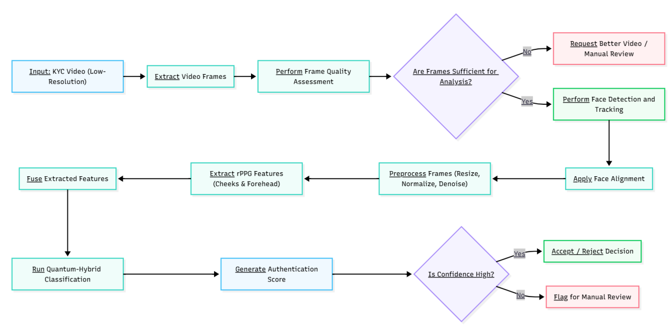

Fig. 1. Proposed Framework for Deepfake Detection in Low-Resolution KYC Systems (visualization created using Mermaid)

As shown in Fig. 1, the proposed system follows a two-stage pipeline.

A. Stage 1: rPPG-Based Physiological Analysis

The system extracts facial regions from video frames and analyzes subtle color variations caused by blood flow using rPPG techniques. These temporal signals capture biological rhythms such as pulse consistency, which are difficult to replicate in deepfake videos.

TABLE I
LITERATURE COMPARISON

Ref	Author	Year	Method/Approach	Findings	Limitations
[1]	Aryaman Dass et al.	2024	Hybrid quantum-classical model using CNN and quantum circuits	Achieved high ROC-AUC (0.995) and strong classification performance	Depends on simulated quantum environment
[2]	Kavya Sree Kammari et al.	2023	rPPG-based physiological signal analysis with ML	Achieved 85–95% accuracy; strong robustness in detecting biological inconsistencies	Sensitive to noise, lighting, and motion
[3]	Xiang Li et al.	2024	Multimodal audio-visual detection using Vision Transformer and temporal attention	Achieved 99.34% accuracy and high AUC score	High computational cost; lacks real-time applicability
[4]	Karan R Talwalkar et al.	2024	Hybrid CNN + GRU model for spatial-temporal feature extraction	Achieved 92.48% accuracy using DenseNet121 + GRU	Limited generalization; lacks real-world KYC focus
[5]	A. Susi et al.	2023	rPPG synchronization with deep learning	Achieved 94% accuracy on Celeb-DF dataset	Sensitive to environmental variations
[6]	Farhaan Khan et al.	2026	Quantum-hybrid deep learning with attention mechanisms	Improved precision and recall in detection	Relies on simulated quantum systems
[7]	Tabish Rao et al.	2024	Multimodal Vision Transformer using spatial, spectral, physiological data	Achieved 98.1% accuracy	Computationally expensive
[8]	Ashifur Rahman et al.	2022	CNN-based detection using MobileNet	Achieved 94.9% accuracy	Limited robustness and feature diversity

TABLE II
COMPARATIVE ANALYSIS

Technique	Advantages	Limitations	Performance
CNN-based Models	Simple, fast, widely used	Cannot capture temporal features	Moderate (85–94%)
CNN + GRU (Hybrid)	Captures spatial and temporal patterns	Inconsistent improvement across models	High (92%)
rPPG-based Methods	Uses physiological signals, better generalization	Sensitive to noise, lighting, motion	High (85–95%)
Multimodal (Audio + Visual)	High accuracy, better robustness	High computational cost	Very High (98–99%)
Transformer-based Models	Captures long-range dependencies	Resource intensive, not real-time	Very High (98–99%)
Quantum Hybrid Models	Potential for advanced classification	Experimental, simulated environments	Promising but not validated

B. Stage 2: Hybrid Quantum Machine Learning Classification

The extracted physiological features are converted into compact feature vectors and passed to a hybrid quantum-classical classifier. This stage performs high-level discrimination between real and fake samples by learning subtle patterns in noisy, low-resolution data.

C. System Workflow

The overall workflow of the proposed system is as follows:

- Video input acquisition (low-resolution KYC video)
- Frame extraction and quality assessment
- Face detection, tracking, and alignment
- Low-resolution preprocessing and normalization
- ROI extraction (cheeks and forehead)
- rPPG signal extraction and feature generation
- Feature conditioning and normalization
- Hybrid quantum classification
- Confidence-based decision output (real/fake/uncertain)

D. Key Advantages

- Handles low-resolution real-world data effectively
- Uses physiological signals for improved reliability
- Integrates quantum learning for advanced classification
- Provides confidence-aware outputs for practical deployment

VI. CHALLENGES AND RESEARCH GAPS

Despite significant advancements, several challenges remain in the domain of deepfake detection.

One of the major limitations is the lack of robustness in real-world scenarios. Most existing models are trained on high-quality datasets and fail to generalize effectively to low-resolution, compressed, or noisy videos commonly encountered in real-world applications such as KYC verification systems.

Another critical gap is the limited use of physiological signals in combination with other modalities. While rPPG-based methods show promise, they are rarely integrated with multimodal or transformer-based architectures for enhanced performance.

Furthermore, existing approaches do not provide confidence-based outputs, which are essential for practical deployment in sensitive applications such as financial verification systems.

Computational complexity is also a significant concern, particularly for transformer-based and multimodal models, which makes real-time deployment challenging.

Additionally, quantum-enhanced models remain largely theoretical and lack validation on real quantum hardware, limiting their practical applicability.

VII. Future Scope

Future research in deepfake detection should focus on developing robust models capable of handling low-resolution and compressed video data. Integrating multimodal approaches with physiological signals such as rPPG can further improve detection accuracy and reliability.

Additionally, lightweight architectures should be explored for real-time deployment in practical systems. Advancements in quantum computing may also enhance detection performance, provided practical implementation becomes feasible.

VIII. Conclusion

This paper presented a comprehensive review of deepfake detection techniques, covering spatial-temporal models, physiological approaches, and multimodal frameworks. While existing methods demonstrate high performance under controlled conditions, they face significant challenges in real-world applications, particularly in low-resolution KYC environments.

To address these limitations, a hybrid framework combining rPPG-based physiological analysis with quantum machine learning classification was proposed. This approach enhances robustness by leveraging biological consistency along with advanced decision-making models.

Future work should focus on improving real-time performance, reducing computational complexity, and validating quantum-enhanced models in practical deployment scenarios. The integration of multimodal and physiological approaches represents a promising direction for building secure and scalable deepfake detection systems.

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